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OPTIMIZATION OF SUPPRESSION OF TRANSVERSE MODE SPURIOUS SIGNALS IN SURFACE ACOUSTIC WAVE FILTERS WHILE MAINTAINING FILTER INSERTION LOSS

Abstract

Aspects and embodiments disclosed herein include a radio frequency ladder filter comprising a plurality of series arm surface acoustic wave resonators electrically connected in series between an input of the filter and an output of the filter, and a plurality of shunt surface acoustic wave resonators each electrically connected between nodes between adjacent one of the plurality of series arm surface acoustic wave resonators and ground, one or more of the plurality of series arm surface acoustic wave resonators including gap hammer structures, a remaining one or more of the plurality of series arm surface acoustic wave resonators lacking gap hammer structures to improve insertion loss characteristics of the filter.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority under 35 U.S.C. § 119 (e) to U.S. Provisional Patent Application Ser. No. 63/560,141, titled “OPTIMIZATION OF SUPPRESSION OF TRANSVERSE MODE SPURIOUS SIGNALS IN SURFACE ACOUSTIC WAVE FILTERS WHILE MAINTAINING FILTER INSERTION LOSS,” filed Mar. 1, 2024, the entire content of which is incorporated herein by reference for all purposes.

BACKGROUND

Technical Field

[0002] Embodiments of this disclosure relate to radio frequency filters including acoustic wave devices and to suppression of transverse mode spurious signals and improvement in insertion loss in same.

Description of Related Technology

[0003] Acoustic wave devices, for example, surface acoustic wave (SAW) and bulk acoustic wave (BAW) devices may be utilized as components of filters in radio frequency electronic systems. For instance, filters in a radio frequency front-end of a mobile phone can include acoustic wave filters. Two acoustic wave filters can be arranged as a duplexer.

SUMMARY

[0004] In accordance with one aspect, there is provided a radio frequency ladder filter. The radio frequency ladder filter comprises a plurality of series arm surface acoustic wave resonators electrically connected in series between an input of the filter and an output of the filter, and a plurality of shunt surface acoustic wave resonators each electrically connected between nodes between adjacent one of the plurality of series arm surface acoustic wave resonators and ground, one or more of the plurality of series arm surface acoustic wave resonators including gap hammer structures, a remaining one or more of the plurality of series arm surface acoustic wave resonators lacking gap hammer structures to improve insertion loss characteristics of the filter.

[0005] In some embodiments, one or more of the plurality of shunt surface acoustic wave resonators include gap hammer structures.

[0006] In some embodiments, each of the plurality of shunt surface acoustic wave resonators include gap hammer structures.

[0007] In some embodiments, one of the plurality of series arm surface acoustic wave resonators has a resonance frequency lower than another of the plurality of series arm surface acoustic wave resonators, the one of the plurality of series arm surface acoustic wave resonators including the gap hammer structure, the another of the plurality of series arm surface acoustic wave resonators lacking the gap hammer structure.

[0008] In some embodiments, the plurality of series arm surface acoustic wave resonators have a respective plurality of resonance frequencies, at least one of the plurality of series arm surface acoustic wave resonators having a lower resonance frequency than at least one other of the plurality of series arm surface acoustic wave resonators including the gap hammer structure.

[0009] In some embodiments, a series arm surface acoustic wave resonator having a lowest resonance frequency among the plurality of series arm surface acoustic wave resonators includes the gap hammer structure.

[0010] In some embodiments, a series arm surface acoustic wave resonator having a highest resonance frequency among the plurality of series arm surface acoustic wave resonators lacks the gap hammer structure.

[0011] In some embodiments, a plurality of series arm surface acoustic wave resonators having

lower resonance frequencies than a series arm surface acoustic wave resonators having a highest resonance frequency among the plurality of series arm surface acoustic wave resonators include the gap hammer structure.

[0012] In accordance with another aspect, there is provided an electronics module comprising a radio frequency ladder filter including a plurality of series arm surface acoustic wave resonators electrically connected in series between an input of the filter and an output of the filter, and a plurality of shunt surface acoustic wave resonators each electrically connected between nodes between adjacent one of the plurality of series arm surface acoustic wave resonators and ground, one or more of the plurality of series arm surface acoustic wave resonators including gap hammer structures, a remaining one or more of the plurality of series arm surface acoustic wave resonators lacking gap hammer structures to improve insertion loss characteristics of the filter.

[0013] In some embodiments, one or more of the plurality of shunt surface acoustic wave resonators include gap hammer structures.

[0014] In some embodiments, each of the plurality of shunt surface acoustic wave resonators include gap hammer structures.

[0015] In some embodiments, one of the plurality of series arm surface acoustic wave resonators has a resonance frequency lower than another of the plurality of series arm surface acoustic wave resonators, the one of the plurality of series arm surface acoustic wave resonators including the gap hammer structure, the another of the plurality of series arm surface acoustic wave resonators lacking the gap hammer structure.

[0016] In some embodiments, the plurality of series arm surface acoustic wave resonators have a respective plurality of resonance frequencies, at least one of the plurality of series arm surface acoustic wave resonators having a lower resonance frequency than at least one other of the plurality of series arm surface acoustic wave resonators including the gap hammer structure.

[0017] In some embodiments, a series arm surface acoustic wave resonator having a lowest resonance frequency among the plurality of series arm surface acoustic wave resonators includes the gap hammer structure.

[0018] In some embodiments, a series arm surface acoustic wave resonator having a highest resonance frequency among the plurality of series arm surface acoustic wave resonators lacks the gap hammer structure.

[0019] In some embodiments, a plurality of series arm surface acoustic wave resonators having lower resonance frequencies than a series arm surface acoustic wave resonator having a highest resonance frequency among the plurality of series arm surface acoustic wave resonators include the gap hammer structure.

[0020] In accordance with another aspect, there is provided a method of forming a radio frequency ladder filter. The method comprises forming a plurality of surface acoustic wave resonators, one or more of the surface acoustic wave resonators including a gap hammer structure, one or more other of the surface acoustic wave resonators lacking a gap hammer structure, electrically connecting a first subset of the plurality of surface acoustic wave resonators in series between an input and an output of the filter, at least one of the first subset of the plurality of surface acoustic wave resonators lacking the gap hammer structure, and electrically connecting a second subset of the plurality of surface acoustic wave resonators between nodes between adjacent one of the first subset of the plurality of series arm surface acoustic wave resonators and ground.

[0021] In some embodiments, at least one of the second subset of the plurality of surface acoustic wave resonators includes a gap hammer structure.

[0022] In some embodiments, at least one of the first subset of the plurality of surface acoustic wave resonators having a resonance frequency lowest among the first subset of the plurality of surface acoustic wave resonators includes a gap hammer structure.

[0023] In some embodiments, at least one of the first subset of the plurality of surface acoustic

wave resonators having a resonance frequency highest among the first subset of the plurality of surface acoustic wave resonators lacks a gap hammer structure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] Embodiments of this disclosure will now be described, by way of non-limiting example, with reference to the accompanying drawings.

[0025] FIG. 1A is a simplified plan view of an example of a surface acoustic wave resonator;

[0026] FIG. 1B is a simplified plan view of another example of a surface acoustic wave resonator;

[0027] FIG. 1C is a simplified plan view of another example of a surface acoustic wave resonator;

[0028] FIG. 2A is a plan view of a portion of electrodes of a surface acoustic wave filter including a structure for suppressing transverse mode spurious signals;

[0029] FIG. 2B is a cross-sectional view of a portion of electrodes of a surface acoustic wave resonator including a structure for suppressing transverse mode spurious signals;

[0030] FIG. 3 illustrates transverse spurious mode signals generated in a SAW resonator including IDT electrodes with a molybdenum layer;

[0031] FIG. 4 illustrates transverse spurious mode signals generated in a SAW resonator including IDT electrodes with a tungsten layer;

[0032] FIG. 5 illustrates an example of IDT electrode extensions in a gap region of a SAW resonator;

[0033] FIG. 6 illustrates another example of IDT electrode extensions in a gap region of a SAW resonator;

[0034] FIG. 7 illustrates a comparison between the insertion loss curves of ladder filters formed with SAW resonators as shown in partial view in FIG. 2A (without IDT electrode extensions) and SAW resonators as shown in partial view in FIG. 5 (with IDT electrode extensions);

[0035] FIG. 8A illustrates a ladder filter formed of SAW resonators in which some of the SAW resonators include IDT electrode extensions (“gap hammers”) and some do not;

[0036] FIG. 8B illustrates a comparison between the insertion loss curves of ladder filters formed with SAW resonators as shown in partial view in FIG. 2A (without IDT electrode extensions) and a ladder filter as shown in FIG. 8A;

[0037] FIG. 9A illustrates another ladder filter formed of SAW resonators in which some of the SAW resonators include IDT electrode extensions (“gap hammers”) and some do not;

[0038] FIG. 9B illustrates a comparison between the insertion loss curves of a ladder filter as shown in FIG. 8A and a ladder filter as shown in FIG. 9A;

[0039] FIG. 10A illustrates another ladder filter formed of SAW resonators in which some of the SAW resonators include IDT electrode extensions (“gap hammers”) and some do not;

[0040] FIG. 10B illustrates another ladder filter formed of SAW resonators in which some of the SAW resonators include IDT electrode extensions (“gap hammers”) and some do not;

[0041] FIG. 10C illustrates another ladder filter formed of SAW resonators in which some of the SAW resonators include IDT electrode extensions (“gap hammers”) and some do not;

[0042] FIG. 11 is a block diagram of one example of a filter module that can include one or more surface acoustic wave filters according to aspects of the present disclosure;

[0043] FIG. 12 is a block diagram of one example of a front-end module that can include one or more filter modules according to aspects of the present disclosure; and

[0044] FIG. 13 is a block diagram of one example of a wireless device including the front-end module of FIG. 12.

DETAILED DESCRIPTION

[0045] The following description of certain embodiments presents various descriptions of specific

embodiments. However, the innovations described herein can be embodied in a multitude of different ways, for example, as defined and covered by the claims. In this description, reference is made to the drawings where like reference numerals can indicate identical or functionally similar elements. It will be understood that elements illustrated in the figures are not necessarily drawn to scale. Moreover, it will be understood that certain embodiments can include more elements than illustrated in a drawing and/or a subset of the elements illustrated in a drawing. Further, some embodiments can incorporate any suitable combination of features from two or more drawings. [0046] FIG. 1A is a plan view of a surface acoustic wave (SAW) resonator **10** such as might be used in a SAW filter, duplexer, etc.

[0047] Acoustic wave resonator **10** is formed on a substrate **12** including a piezoelectric material layer, for example, a lithium tantalate (LiTaO₃) or lithium niobate (LiNbO₃) material layer. In some embodiments the substrate **12** may be a multilayer piezoelectric substrate (MPS). The acoustic wave resonator **10** includes Interdigital Transducer (IDT) electrodes **14** and reflector electrodes **16**. In use, the IDT electrodes **14** excite a main acoustic wave having a wavelength λ along a surface of the substrate **12**. The reflector electrodes **16** sandwich the IDT electrodes **14** and reflect the main acoustic wave back and forth through the IDT electrodes **14**. The main acoustic wave of the device travels perpendicular to the lengthwise direction of the IDT electrodes.

[0048] The IDT electrodes **14** include a first bus bar electrode **18A** and a second bus bar electrode **18B** facing the first bus bar electrode **18A**. The IDT electrodes **14** further include first electrode fingers **20A** extending from the first bus bar electrode **18A** toward the second bus bar electrode **18B**, and second electrode fingers **20B** extending from the second bus bar electrode **18B** toward the first bus bar electrode **18A**.

[0049] The reflector electrodes **16** (also referred to as reflector gratings) each include a first reflector bus bar electrode **24A** and a second reflector bus bar electrode **24B** and reflector fingers **26** extending between and electrically coupling the first bus bar electrode **24A** and the second bus bar electrode **24B**.

[0050] In other embodiments disclosed herein, as illustrated in FIG. 1B, the reflector bus bar electrodes **24A**, **24B** may be omitted and the reflector fingers **26** may be electrically unconnected. Further, as illustrated in FIG. 1C, acoustic wave resonators as disclosed herein may include dummy electrode fingers **20C** that are aligned with respective electrode fingers **20A**, **20B**. Each dummy electrode finger **20C** extends from the opposite bus bar electrode **18A**, **18B** than the respective electrode finger **20A**, **20B** with which it is aligned.

[0051] It should be appreciated that the acoustic wave resonators **10** illustrated in FIGS. 1A-1C, as well as the other circuit elements illustrated in other figures presented herein, are illustrated in a highly simplified form. The relative dimensions of the different features are not shown to scale. Further, typical acoustic wave resonators would commonly include a far greater number of electrode fingers and reflector fingers than illustrated. Typical acoustic wave resonators or filter elements may also include multiple IDT electrodes sandwiched between the reflector electrodes.

[0052] As illustrated in FIG. 2A, regions along lengths of the IDT electrodes of a SAW device, e.g., a SAW resonator, may be characterized as busbar regions “B” including the busbar portions of the IDT electrodes, gap regions “G” between the busbar of a first set of IDT electrodes and the ends of the fingers of a second set of IDT electrode extending from a second busbar of the SAW device, edge regions “E” including end portions of the IDT electrodes, and a center region “C” sandwiched between the edge regions. In some embodiments, the gap regions may have widths of between about 1λ and 1.5λ , the edge regions may have widths of between about 0.25λ and 1.25λ , and the center region may have a width of about 20λ , although it should be understood that these dimensions are only examples and may vary based on the design of a particular resonator. In some embodiments, a layer of a dielectric **22** exhibiting a high acoustic wave velocity, for example, silicon nitride (Si₃N₄, also abbreviated as “SiN” herein) may be disposed over the IDT electrodes within the center region C. In some embodiments, as illustrated in FIG. 2B, the layer of

high acoustic wave velocity material **22** may be deposited over a dielectric material **32** having a lower acoustic wave velocity, for example, silicon dioxide (SiO₂) disposed over the entire IDT electrode structure (regions B, G, E, and D). The layer of high acoustic wave velocity material **22** may include a thicker portion disposed in the center region C than in the other regions B, G, and E. The layer of high acoustic wave velocity material **22** disposed over the IDT electrodes in the center region C may help to confine acoustic waves to the center region C and reduce the amount of acoustic energy that travels outside of this region in a direction perpendicular to that of the propagation direction of the main acoustic wave in the device and that may cause transverse mode spurious signals in the frequency response of the SAW device. The main acoustic wave of the device travels perpendicular to the lengthwise direction of the IDT electrodes and the transverse mode spurious signals may be caused by acoustic waves travelling parallel to the lengthwise direction of the IDT electrodes.

[0053] The SiO₂ layer **32** may have a negative temperature coefficient of frequency, which helps to offset the positive temperature coefficient of frequency of the piezoelectric substrate **12** and reduce the change in frequency response of the SAW device with changes in temperature. A SAW device with a layer of SiO₂ over the IDT electrodes may thus be referred to as a temperature-compensated SAW device, or TCSAW.

[0054] As also illustrated in FIG. 2B, the IDT electrodes **14** may be layered electrodes including an upper layer **14A** of a highly conductive but low-density material, for example, aluminum (Al), and a lower layer **14B** of a less conductive, but more dense material, for example, molybdenum (Mo) or tungsten (W). The denser lower layer **14B** may reduce the acoustic velocity of acoustic waves travelling through the device which may allow the IDT electrode fingers to be spaced more closely for a given operating frequency and allow the SAW device to be reduced in size as compared to a similar device utilizing less dense IDT electrodes. The less dense upper layer **14A** may have a higher conductivity than the lower layer **14B** to provide the electrode stack with a lower overall resistivity.

[0055] Consumers and device manufacturers continue to demand electronic products such as cellular telephones with smaller form factors and/or that include additional functionality. Accordingly, there is a continuing demand for smaller and smaller electronic components used in these electronic products, for example, SAW resonators and filters that are incorporated in same. A method of decreasing the size of a SAW resonator while maintaining the operating frequency of the SAW resonator includes increasing the density of the IDT electrodes of the SAW resonator. Higher density IDT electrodes result in a reduced velocity and reduced wavelength of acoustic waves generated in the SAW resonator, which allows denser IDT electrodes to be spaced closer to each other than less dense IDT electrodes to achieve the same operating frequency. In many examples of previously and currently available SAW resonators, the IDT electrodes were formed of Mo, often with a layer of Al on top of the Mo to increase the conductivity of the IDT electrodes. To increase the density of the IDT electrodes, the Mo layer of the IDT electrodes may be replaced with a layer of a higher density material, for example, W.

[0056] It has been discovered that when the Mo layer in IDT electrodes of examples of SAW resonators is replaced with W, previously utilized structures, such as the layer of high acoustic wave velocity material **22** disposed over the IDT electrodes in their center region C, may be less effective than desirable in suppressing transverse mode spurious signals that may interfere with operation of the SAW resonator. FIG. 3 illustrates the strength of transverse mode spurious signals generated in a SAW resonator utilizing IDT electrodes with a Mo layer and a silicon nitride layer thickness of 0.005λ , where λ represents the wavelength of the main acoustic wave generated in the resonator. In comparison, FIG. 4 illustrates the strength of transverse mode spurious signals generated in a similar SAW resonator utilizing IDT electrodes with a W layer and various silicon nitride layer thickness. It can be seen that even if the thickness of the silicon nitride layer is more than doubled in the SAW resonator utilizing the IDT electrodes with the W layer as compared to

that of the SAW resonator utilizing the IDT electrodes with the Mo layer, the transverse mode spurious signals generated in the SAW resonator utilizing the IDT electrodes with the W layer are significantly stronger than the transverse mode spurious signals generated in the SAW resonator utilizing the IDT electrodes with the Mo layer.

[0057] One method of reducing the strength of transverse mode spurious signals in a SAW resonator may be to include additional material, for example, extensions of the IDT electrodes in the gap regions of the SAW resonator. FIG. 5 illustrates one example of a SAW resonator including extensions **20C** of the IDT electrodes in the gap regions of the resonator. Only the region proximate one of the busbars **18A** and extensions **20C** from only one set **20A** of IDT electrodes is illustrated. It is to be appreciated that the same form of IDT electrode extensions **20C** may be present in the gap region on the unillustrated side of the resonator extending from the second set **20B** of IDT electrodes (See FIG. 6 below). The resonator may include the layer of high acoustic wave velocity material **22** over the IDT electrodes in the center region of the resonator and, in some embodiments, the layer of dielectric material **32** below the layer of material **22** as well. The IDT electrode extensions **20C** may be referred to herein as “gap hammers.”

[0058] The IDT electrode extensions **20C** in the gap region of the resonator may be in the form of rectangles of the same material or materials that the IDT electrodes are formed of that extend perpendicular to the lengthwise direction of the IDT electrodes. In some embodiments, the IDT electrode extensions **20C** from one set of IDT electrodes **20A** may extend to a position overlapping an extending region **21** of tips **20D** of the other set of IDT electrodes **20B** to define an overlap region **28**. The extending region **21** may be defined by lines extending into the gap region from opposite lengthwise sides of the other set of IDT electrodes **20B** in the edge region (and on the other side of the resonator, from the sides of the set of IDT electrodes **20A** in the edge region).

[0059] In some embodiments, the IDT electrode extensions **20C** may extend through and terminate at ends of the extending regions **21** and overlap an entirety of the widths of the tips **20D** of the other set of IDT electrodes **20B**, in other embodiments, may extend only partially through the extending regions **21** and only partially overlap the widths of the tips **20D** of the other set of IDT electrodes **20B**, and in other embodiments, the IDT electrode extensions **20C** may extend beyond the edges of the extension regions **21** and beyond the tips **20D** of the other set of IDT electrodes **20B**. Unless specified otherwise, as the term is used herein, a width direction is in a direction parallel to the direction of propagation of the main acoustic wave through the SAW resonator. Unless specified otherwise, as the term is used herein, a lengthwise direction is perpendicular to the direction of propagation of the main acoustic wave through the SAW resonator.

[0060] Although illustrated as rectangular structures extending from only a single side of the IDT electrodes **20A** in FIG. 5, in other embodiments, the IDT electrode extensions **20C** may extend from both sides of the IDT electrodes **20A** (and/or IDT electrodes **20B**). It should be appreciated that the IDT electrode extensions **20C** are not limited to having a rectangular shape. In other embodiments, the IDT electrode extensions **20C** may be square, oval, circular, or may have any other shape desired.

[0061] The gap hammers **20C** may have lengthwise dimensions (a) as shown in FIG. 6 that may be between 0.10λ and 0.30λ . The gap hammers **20C** may be spaced from tips of opposing electrode fingers by a distance (b) of between 0.10λ and 0.30λ . In some embodiments, lengths (a) and (b) are equal so that a “gap hammer duty factor” of $a/(a+b)=0.50$.

[0062] As noted above the gap hammers **20C** may help suppress transverse mode spurious signals in the admittance curve of a SAW resonator. However, the gap hammers **20C** may also reduce the quality factor of the resonator, which may manifest as deterioration of the insertion loss curve for a ladder filter formed of SAW resonators including the gap hammers. FIG. 7 illustrates a comparison between the measured insertion loss curve of a ladder filter formed of SAW resonators without gap hammers such as illustrated in part in FIG. 2A and a ladder filter formed of SAW resonators including gap hammers such as illustrated in part in FIG. 5. It can be observed that the ladder filter

formed of the SAW resonators including the gap hammers exhibits a slight degradation in insertion loss at the upper end of the passband.

[0063] To at least partially alleviate the degradation in insertion loss of a ladder filter due to the presence of the gap hammers in the SAW resonators forming the filter while retaining at least some, if not most, of the benefit of reduced spurious signals in resonator admittance curves due to the gap hammers, one may form the ladder filter with some of the SAW resonators including the gap hammers and some of the SAW resonators not including the gap hammers, for example, with some of the resonators having IDT structures as illustrated in part in FIG. 2A and others of the resonators including IDT structures as illustrated in FIG. 5.

[0064] FIG. 8A illustrates one example of a ladder filter (“B7 Filter Topology v1”) including four series arm resonators and four shunt resonators. Two out of four of the series arm resonators include gap hammer structures (“With GapH”) while the other two do not (“No GapH”). Three out of four of the shunt resonators include gap hammer structures while one does not. FIG. 8B compares the measured insertion loss curve for the ladder filter of FIG. 8A against a ladder filter with a similar layout but in which all resonators lack gap hammer structures. The insertion loss curve for the “B7 Filter Topology v1” ladder filter is improved as compared to that of the ladder filter with all resonators lacking gap hammer structures.

[0065] It has been discovered that even further improvements in filter insertion loss may be achieved if the series arm resonators including the gap hammer structures were selected from the series arm resonators with lower resonance frequencies and series arm resonators without the gap hammer structures were selected from the series arm resonators with higher resonance frequencies. As illustrated in FIG. 9A a ladder filter (“B7 Filter Topology v2”) includes series arm resonators S1, S2, S3, S4, each with different resonance frequencies. Resonator S1 has the lowest resonance frequency, S4 has the second lowest resonance frequency, S2 the second highest resonance frequency, and S3 has the highest resonance frequency. Gap hammer structures are included in the two series arm resonators with the lower resonance frequencies, namely S1 and S4, while the two series arm resonators with the higher resonance frequencies, namely S2 and S3 do not include gap hammer structures. Shunt resonators P1, P2, and P3 include gap hammer structures while shunt resonator P4 does not.

[0066] FIG. 9B compares the measured insertion loss curve of the ladder filter with the v2 topology as illustrated in FIG. 9A against the ladder filter with the v1 topology as illustrated in FIG. 8A. It can be seen that the ladder filter with the v2 topology as illustrated in FIG. 9A exhibits an improved insertion loss curve, especially on the high frequency side, as compared to the ladder filter with the v1 topology as illustrated in FIG. 8A.

[0067] It should be noted that the positions of the series arm resonators with the higher and lower resonance frequencies need not be as shown in FIG. 9A. Any of the series arm resonators may have the highest resonance frequency, any other of the series arm resonators may have the second highest resonance frequency, any other of the series arm resonators may have the second lowest resonance frequency, and any other of the series arm resonators may have the lowest resonance frequency among the group of series arm resonators. Furthermore, in different embodiments, a ladder filter may include a different number of series arm resonators with gap hammer structures. As an example, FIG. 10A illustrates a ladder filter in which only the series arm resonator S1 with the lowest resonance frequency among the series arm resonators includes a gap hammer structure. FIG. 10B illustrates an embodiment in which all series arm resonators except the one having the highest resonance frequency, namely S1, include gap hammer structures. In other embodiments, any of the shunt resonators may include or not include gap hammer structures. FIG. 10C illustrates an example of a ladder filter similar to that of FIG. 9A but in which all of the shunt resonators include gap hammer structures. In further embodiments, ladder filters may include more or fewer series arm and/or shunt resonators than illustrated in the examples depicted herein.

[0068] Filter including SAW resonators with and without gap hammer structures as discussed

herein can be implemented in a variety of packaged modules. Some example packaged modules will now be discussed in which any suitable principles and advantages of the filters discussed herein can be implemented. FIGS. **11**, **12**, and **13** are schematic block diagrams of illustrative packaged modules and devices according to certain embodiments.

[0069] As discussed above, embodiments of a surface acoustic wave (SAW) filter using one or more surface acoustic wave elements may be incorporated into and packaged as a module that may ultimately be used in an electronic device, such as a wireless communications device, for example. FIG. **11** is a block diagram illustrating one example of a module **300** including a SAW filter **310**. The SAW filter **310** may be implemented on one or more die(s) **320** including one or more connection pads **322**. For example, the SAW filter **310** may include a connection pad **322** that corresponds to an input contact for the SAW filter and another connection pad **322** that corresponds to an output contact for the SAW filter. The packaged module **300** includes a packaging substrate **330** that is configured to receive a plurality of components, including the die **320**. A plurality of connection pads **332** can be disposed on the packaging substrate **330**, and the various connection pads **322** of the SAW filter die **320** can be connected to the connection pads **332** on the packaging substrate **330** via electrical connectors **334**, which can be solder bumps or wirebonds, for example, to allow for passing of various signals to and from the SAW filter **310**. The module **300** may optionally further include other circuitry die **340**, for example, one or more additional filter(s), amplifiers, pre-filters, modulators, demodulators, down converters, and the like, as would be known to one of skill in the art of semiconductor fabrication in view of the disclosure herein. In some embodiments, the module **300** can also include one or more packaging structures to, for example, provide protection and facilitate easier handling of the module **300**. Such a packaging structure can include an overmold formed over the packaging substrate **330** and dimensioned to substantially encapsulate the various circuits and components thereon.

[0070] Various examples and embodiments of the SAW filter **310** can be used in a wide variety of electronic devices. For example, the SAW filter **310** can be used in an antenna duplexer, which itself can be incorporated into a variety of electronic devices, such as RF front-end modules and communication devices.

[0071] Referring to FIG. **12**, there is illustrated a block diagram of one example of a front-end module **400**, which may be used in an electronic device such as a wireless communications device (e.g., a mobile phone) for example. The front-end module **400** includes an antenna duplexer **410** having a common node **402**, an input node **404**, and an output node **406**. An antenna **510** is connected to the common node **402**.

[0072] The antenna duplexer **410** may include one or more transmission filters **412** connected between the input node **404** and the common node **402**, and one or more reception filters **414** connected between the common node **402** and the output node **406**. The passband(s) of the transmission filter(s) are different from the passband(s) of the reception filters. Examples of the SAW filter **310** can be used to form the transmission filter(s) **412** and/or the reception filter(s) **414**. An inductor or other matching component **420** may be connected at the common node **402**.

[0073] The front-end module **400** further includes a transmitter circuit **432** connected to the input node **404** of the duplexer **410** and a receiver circuit **434** connected to the output node **406** of the duplexer **410**. The transmitter circuit **432** can generate signals for transmission via the antenna **510**, and the receiver circuit **434** can receive and process signals received via the antenna **510**. In some embodiments, the receiver and transmitter circuits are implemented as separate components, as shown in FIG. **12**, however in other embodiments these components may be integrated into a common transceiver circuit or module. As will be appreciated by those skilled in the art, the front-end module **400** may include other components that are not illustrated in FIG. **12** including, but not limited to, switches, electromagnetic couplers, amplifiers, processors, and the like.

[0074] FIG. **13** is a block diagram of one example of a wireless device **500** including the antenna duplexer **410** shown in FIG. **12**. The wireless device **500** can be a cellular phone, smart phone,

tablet, modem, communication network or any other portable or non-portable device configured for voice or data communication. The wireless device **500** can receive and transmit signals from the antenna **510**. The wireless device includes an embodiment of a front-end module **400** similar to that discussed above with reference to FIG. **12**. The front-end module **400** includes the duplexer **410**, as discussed above. In the example shown in FIG. **13** the front-end module **400** further includes an antenna switch **440**, which can be configured to switch between different frequency bands or modes, such as transmit and receive modes, for example. In the example illustrated in FIG. **13**, the antenna switch **440** is positioned between the duplexer **410** and the antenna **510**; however, in other examples the duplexer **410** can be positioned between the antenna switch **440** and the antenna **510**. In other examples the antenna switch **440** and the duplexer **410** can be integrated into a single component.

[0075] The front-end module **400** includes a transceiver **430** that is configured to generate signals for transmission or to process received signals. The transceiver **430** can include the transmitter circuit **432**, which can be connected to the input node **404** of the duplexer **410**, and the receiver circuit **434**, which can be connected to the output node **406** of the duplexer **410**, as shown in the example of FIG. **12**.

[0076] Signals generated for transmission by the transmitter circuit **432** are received by a power amplifier (PA) module **450**, which amplifies the generated signals from the transceiver **430**. The power amplifier module **450** can include one or more power amplifiers. The power amplifier module **450** can be used to amplify a wide variety of RF or other frequency-band transmission signals. For example, the power amplifier module **450** can receive an enable signal that can be used to pulse the output of the power amplifier to aid in transmitting a wireless local area network (WLAN) signal or any other suitable pulsed signal. The power amplifier module **450** can be configured to amplify any of a variety of types of signal, including, for example, a Global System for Mobile (GSM) signal, a code division multiple access (CDMA) signal, a W-CDMA signal, a Long-Term Evolution (LTE) signal, or an EDGE signal. In certain embodiments, the power amplifier module **450** and associated components including switches and the like can be fabricated on gallium arsenide (GaAs) substrates using, for example, high-electron mobility transistors (pHEMT) or insulated-gate bipolar transistors (BiFET), or on a Silicon substrate using complementary metal-oxide semiconductor (CMOS) field effect transistors.

[0077] Still referring to FIG. **13**, the front-end module **400** may further include a low noise amplifier module **460**, which amplifies received signals from the antenna **510** and provides the amplified signals to the receiver circuit **434** of the transceiver **430**.

[0078] The wireless device **500** of FIG. **13** further includes a power management sub-system **520** that is connected to the transceiver **430** and manages the power for the operation of the wireless device **500**. The power management system **520** can also control the operation of a baseband sub-system **530** and various other components of the wireless device **500**. The power management system **520** can include, or can be connected to, a battery (not shown) that supplies power for the various components of the wireless device **500**. The power management system **520** can further include one or more processors or controllers that can control the transmission of signals, for example. In one embodiment, the baseband sub-system **530** is connected to a user interface **540** to facilitate various input and output of voice and/or data provided to and received from the user. The baseband sub-system **530** can also be connected to memory **550** that is configured to store data and/or instructions to facilitate the operation of the wireless device, and/or to provide storage of information for the user. Any of the embodiments described above can be implemented in association with mobile devices such as cellular handsets. The principles and advantages of the embodiments can be used for any systems or apparatus, such as any uplink wireless communication device, that could benefit from any of the embodiments described herein. The teachings herein are applicable to a variety of systems. Although this disclosure includes some example embodiments, the teachings described herein can be applied to a variety of structures. Any of the principles and

advantages discussed herein can be implemented in association with RF circuits configured to process signals in a range from about 30 kHz to 5 GHz, such as in a range from about 600 MHz to 2.7 GHz.

[0079] Aspects of this disclosure can be implemented in various electronic devices. Examples of the electronic devices can include, but are not limited to, consumer electronic products, parts of the consumer electronic products such as packaged radio frequency modules, uplink wireless communication devices, wireless communication infrastructure, electronic test equipment, etc. Examples of the electronic devices can include, but are not limited to, a mobile phone such as a smart phone, a wearable computing device such as a smart watch or an ear piece, a telephone, a television, a computer monitor, a computer, a modem, a hand-held computer, a laptop computer, a tablet computer, a microwave, a refrigerator, a vehicular electronics system such as an automotive electronics system, a stereo system, a digital music player, a radio, a camera such as a digital camera, a portable memory chip, a washer, a dryer, a washer/dryer, a copier, a facsimile machine, a scanner, a multi-functional peripheral device, a wrist watch, a clock, etc. Further, the electronic devices can include unfinished products.

[0080] Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise,” “comprising,” “include,” “including” and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to.” The word “coupled”, as generally used herein, refers to two or more elements that may be either directly connected, or connected by way of one or more intermediate elements. Likewise, the word “connected”, as generally used herein, refers to two or more elements that may be either directly connected, or connected by way of one or more intermediate elements. Additionally, the words “herein,” “above,” “below,” and words of similar import, when used in this application, shall refer to this application as a whole and not to any particular portions of this application. Where the context permits, words in the above Detailed Description using the singular or plural number may also include the plural or singular number respectively. The word “or” in reference to a list of two or more items, that word covers all of the following interpretations of the word: any of the items in the list, all of the items in the list, and any combination of the items in the list.

[0081] Moreover, conditional language used herein, such as, among others, “can,” “could,” “might,” “may,” “e.g.,” “for example,” “such as” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or states. Thus, such conditional language is not generally intended to imply that features, elements and/or states are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or states are included or are to be performed in any particular embodiment.

[0082] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel apparatus, methods, and systems described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the methods and systems described herein may be made without departing from the spirit of the disclosure. For example, while blocks are presented in a given arrangement, alternative embodiments may perform similar functionalities with different components and/or circuit topologies, and some blocks may be deleted, moved, added, subdivided, combined, and/or modified. Each of these blocks may be implemented in a variety of different ways. Any suitable combination of the elements and acts of the various embodiments described above can be combined to provide further embodiments. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure.

Claims

- 1.** A radio frequency ladder filter comprising: a plurality of series arm surface acoustic wave resonators electrically connected in series between an input of the filter and an output of the filter; and a plurality of shunt surface acoustic wave resonators each electrically connected between nodes between adjacent one of the plurality of series arm surface acoustic wave resonators and ground, one or more of the plurality of series arm surface acoustic wave resonators including gap hammer structures, a remaining one or more of the plurality of series arm surface acoustic wave resonators lacking gap hammer structures to improve insertion loss characteristics of the filter.
- 2.** The radio frequency ladder filter of claim 1 wherein one or more of the plurality of shunt surface acoustic wave resonators include gap hammer structures.
- 3.** The radio frequency ladder filter of claim 2 wherein each of the plurality of shunt surface acoustic wave resonators include gap hammer structures.
- 4.** The radio frequency ladder filter of claim 1 wherein one of the plurality of series arm surface acoustic wave resonators has a resonance frequency lower than another of the plurality of series arm surface acoustic wave resonators, the one of the plurality of series arm surface acoustic wave resonators including the gap hammer structure, the another of the plurality of series arm surface acoustic wave resonators lacking the gap hammer structure.
- 5.** The radio frequency ladder filter of claim 1 wherein the plurality of series arm surface acoustic wave resonators have a respective plurality of resonance frequencies, at least one of the plurality of series arm surface acoustic wave resonators having a lower resonance frequency than at least one other of the plurality of series arm surface acoustic wave resonators including the gap hammer structure.
- 6.** The radio frequency ladder filter of claim 5 wherein a series arm surface acoustic wave resonator having a lowest resonance frequency among the plurality of series arm surface acoustic wave resonators includes the gap hammer structure.
- 7.** The radio frequency ladder filter of claim 5 wherein a series arm surface acoustic wave resonator having a highest resonance frequency among the plurality of series arm surface acoustic wave resonators lacks the gap hammer structure.
- 8.** The radio frequency ladder filter of claim 5 wherein a plurality of series arm surface acoustic wave resonators having lower resonance frequencies than a series arm surface acoustic wave resonators having a highest resonance frequency among the plurality of series arm surface acoustic wave resonators include the gap hammer structure.
- 9.** An electronics module comprising a radio frequency ladder filter including a plurality of series arm surface acoustic wave resonators electrically connected in series between an input of the filter and an output of the filter, and a plurality of shunt surface acoustic wave resonators each electrically connected between nodes between adjacent one of the plurality of series arm surface acoustic wave resonators and ground, one or more of the plurality of series arm surface acoustic wave resonators including gap hammer structures, a remaining one or more of the plurality of series arm surface acoustic wave resonators lacking gap hammer structures to improve insertion loss characteristics of the filter.
- 10.** The electronics module of claim 9 wherein one or more of the plurality of shunt surface acoustic wave resonators include gap hammer structures.
- 11.** The electronics module of claim 10 wherein each of the plurality of shunt surface acoustic wave resonators include gap hammer structures.
- 12.** The electronics module of claim 9 wherein one of the plurality of series arm surface acoustic wave resonators has a resonance frequency lower than another of the plurality of series arm surface acoustic wave resonators, the one of the plurality of series arm surface acoustic wave resonators including the gap hammer structure, the another of the plurality of series arm surface acoustic wave resonators lacking the gap hammer structure.

resonators lacking the gap hammer structure.

13. The electronics module filter of claim 9 wherein the plurality of series arm surface acoustic wave resonators have a respective plurality of resonance frequencies, at least one of the plurality of series arm surface acoustic wave resonators having a lower resonance frequency than at least one other of the plurality of series arm surface acoustic wave resonators including the gap hammer structure.

14. The electronics module of claim 13 wherein a series arm surface acoustic wave resonator having a lowest resonance frequency among the plurality of series arm surface acoustic wave resonators includes the gap hammer structure.

15. The electronics module of claim 13 wherein a series arm surface acoustic wave resonator having a highest resonance frequency among the plurality of series arm surface acoustic wave resonators lacks the gap hammer structure.

16. The electronics module of claim 13 wherein a plurality of series arm surface acoustic wave resonators having lower resonance frequencies than a series arm surface acoustic wave resonators having a highest resonance frequency among the plurality of series arm surface acoustic wave resonators include the gap hammer structure.

17. A method of forming a radio frequency ladder filter, the method comprising: forming a plurality of surface acoustic wave resonators, one or more of the surface acoustic wave resonators including a gap hammer structure, one or more other of the surface acoustic wave resonators lacking a gap hammer structure; electrically connecting a first subset of the plurality of surface acoustic wave resonators in series between an input and an output of the filter, at least one of the first subset of the plurality of surface acoustic wave resonators lacking the gap hammer structure; and electrically connecting a second subset of the plurality of surface acoustic wave resonators between nodes between adjacent one of the first subset of the plurality of series arm surface acoustic wave resonators and ground.

18. The method of claim 17 wherein at least one of the second subset of the plurality of surface acoustic wave resonators includes a gap hammer structure.

19. The method of claim 17 wherein at least one of the first subset of the plurality of surface acoustic wave resonators having a resonance frequency lowest among the first subset of the plurality of surface acoustic wave resonators includes a gap hammer structure.

20. The method of claim 17 wherein at least one of the first subset of the plurality of surface acoustic wave resonators having a resonance frequency highest among the first subset of the plurality of surface acoustic wave resonators lacks a gap hammer structure.
